

Gaussian and Impulse Noise Identification from Image Using Frequency Domain Analysis



Aakanksha Jain and Harshal Arolkar

Abstract Classifying the various types of noise and reconstructing de-noisy images is one of the challenging issues in the field of image processing. Noise, compression artifacts, blurring, and sensor limits are a few examples of the many causes of distortion. To increase data quality and signal processing in a variety of fields, such as pictures, audio processing, and telecommunications, precise noise type identification is necessary for the development of noise reduction algorithms suited to particular applications. The statistical characteristics of Gaussian noise have a Gaussian (normal) distribution. It is usually enhanced to the overall variability of a signal and frequently linked to environmental factors. On the other hand, abrupt, high-intensity spikes or signal disruptions are characteristic of impulse noise. In this paper, the frequency domain analysis approach is utilized to find Gaussian blur and impulse noise in an image. Signals or time-domain data can be converted into its frequency domain representation using a computing process called the Fourier transform (FFT). This study suggests a two-step method for classifying different types of noise. First, the frequency spectrum is shown by applying the FFT to the noisy input. Then, statistical measures such as mean, standard deviation, and z-score of spectrum are computed. FFT is used to characterize the spectral characteristics of Gaussian and impulsive noise from different images.

Keywords Statistical analysis · Fourier transform · Image noise classification · Frequency domain analysis

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1 Introduction

Reconstructing the noise-free images after categorizing the noises in the images is a difficult perceptual challenge. For classifying picture noise, there are many different approaches and methodologies. Noise in digital images can take many forms, making it essential to identify and categorize these disturbances. The most important factor in images is that it should be noise-free. Image de-noising is necessary in a variety of applications, including text extraction, image categorization, clear image appearance, and license plate detection. It is necessary to identify the type of noise present in an image before de-noising it. There are many techniques for spatial and transform image de-noising and noise categorization.

The Gaussian filter, median filter, and bilateral filter are spatial domain techniques that work directly on the pixel values of the image. Depending on the choice of the basis changes functions, the transform domain filtering techniques can be categorized. Data adaptive and non-data adaptive methods are the parts of transform domain methods. Data adaptive de-noising adjusts its de-noising method based on the characteristics of the input images and uses non-fixed thresholds for noise categorization. The non-data adaptive technique is applied uniformly to all data points, regardless of their individual properties. The statistical measurements of the training images are used to calculate thresholds. Due to the ability to identify patterns and training data characteristics, the transform domain is often used.

2 Literature Review

Liu et al. [1] worked on a convolutional neural network (CNN) model which can identify noise from images. This method can identify Gaussian, salt and pepper, Poisson, and speckle type of noises from the image. Statistical measurements include peak signal to noise ratio (PSNR), and structural similarity index measures are used for calculating performance of the proposed method. Berkeley Segmentation Dataset (BSD500) dataset is used for testing. Noise recognition rates for gray type images are 90.1% and for RGB images are 93.7%.

Khaw et al. [2] proposed a model of a combination of CNN and principal component analysis (PCA). PCA is used as a statistical method to decrease variance of an image. Several standard images used by researchers are such as Lena, cameraman, house, and pepper for testing. The model has accurately identified 86.3% of images.

Momeny et al. [3] proposed a method named noise-robust convolutional neural network (NR-CNN) which can work on specific images like medical CT images or remote images. This method identifies impulse noise from an image. They used statistical methods for finding accuracy of noise classification. CNN gives lower accuracy values in comparison with NR-CNN. Using CNN method classification accuracy varies from 31 to 90%, while NR-CNN gives accuracy above 90% in different test images.

Benesty et al. [4] worked on a convolutional de-noising autoencoder (CDAE) and FFT method for identification of noise. They worked on TIMIT database for testing. And PSNR and SNR evaluation criteria are used to calculate the noise percentage.

Hsieh et al. [5] used the various filters like wiener, bilateral, principal component analysis, and convolutional neural network for identification of whether Gaussian noise is present or not in an image. They used image quality assessment (IQA) dataset for testing. Researchers proposed noise classification on softmax and Visual Geometry Group (VGG16) models of a CNN. Softmax model gave 24.3% validation accuracy to classifying distortion type, while VGG gave better accuracy with 98%.

The median filter and Fourier domain of the Wiener filter are used by Hambal et al. [6] for identifying the noises like speckle, periodic, uniform, or salt and pepper. PSNR is used for evaluating the type of noise. Median filter gives better accuracy.

Wu and Gao [7] proposed a method of combining de-noising convolutional neural network (DnCNN) and residual encoder–decoder networks (REDNet) to identify noise and denoise it. A research gap is obtained from this method that the suppression of troublesome noise and irregular noise from Gaussian noise is still a difficulty.

Win et al. [8] used a nonlinear filter–median filter for noise reduction after identifying the noise type. This method identifies target noise like Gaussian, salt and pepper, or speckle noise. For testing purposes, they took real-time noisy images recorded by a digital camera.

Sil et al. [9] worked on CNN, blind de-noising for identification of Gaussian noise, uniform noise, exponential noise, Poisson noise, salt-and-pepper noise, speckle noise, Rayleigh noise, and log-normal noise. In this paper, CNN does not give much improvement with respect to other types of noises. Wavelet transform is used for classifying Gaussian noise using a quality metric. The BSD300 image dataset is utilized for study.

Kumar and Agarwal [10] used HAAR wavelet transform method for identifying different types of noises like Gaussian, speckle, salt and pepper, and Poisson. For testing purposes, a white flower image has been taken by researchers. They used various statistical techniques like mean, mode, variance, and PSNR in which speckle noise gave the better result using wavelet transform method.

3 Dataset Used in Experiment

The experiment has been carried out on some Pristine images from Konstanz Artificially Distorted Image Quality Database (Kadik 10k). It contains 81 pristine images, each degraded by 25 distortions in 5 levels. Researchers have focused on two types of noise namely Gaussian blur and impulse noise. The Google Colab environment is used for the experiment. Figure 1 shows two original images along with Gaussian blur and impulse noise taken from Kadik 10k dataset.

The first column displays the original images, and the second and third columns showcase the Gaussian and impulsive noise versions of the same images, respectively.



Fig. 1 Sample images from Kadik10K dataset

4 Methodology

Frequency domain analysis is used to categorize two different types of noise namely Gaussian blur and impulse noise. Noise can be reduced by filtering particular frequency components that are linked to noise. Gaussian blur noise results from image compression, resizing, or optical issues in capturing devices. The impulse noise appears as random, bright, or dark pixels scattered throughout the image which results from data transmission, sensor malfunctions, and other forms of interference.

In the frequency domain, Gaussian blur noise frequently decreases the high-frequency components of an image, resulting in a smoother frequency range. So, frequency domain analysis can be used to find noise from images because high-frequency components are reduced.

Impulse noise, commonly referred to as salt-and-pepper noise, arises when pixel values are randomly corrupted. The smooth transition between frequencies can be disturbed in the frequency domain by impulse noise, which can add random spikes to the frequency spectrum. When pixel values are compared to their surrounding values, outliers that differ significantly can be used to detect impulse noise. Flow of the entire methodology is shown in Fig. 2.

4.1 Algorithm

The algorithm used in the experiment is as mentioned.

1. Load and preprocess train and test images

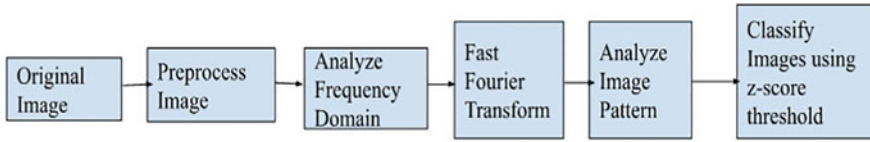


Fig. 2 General flow of experiment

2. For each trained image:

- Compute magnitude spectrum (FFT).

$$F[u, v] = \sum \sum (f(x, y) * \exp(-j * 2 * \pi * (ux/N + vy/M)));$$

for $x = 0, 1, \dots, N - 1$; $y = 0, 1, \dots, M - 1$

- Calculate mean magnitude and standard deviation of image.

$$\text{Mean Magnitude } m = (\sum I[i, j]) / (r * c)$$

$$\text{Standard Deviation} = \sqrt{((\sum (I[i, j] - m)^2) / (r * c))}$$

(where, an image represented by a matrix I where each element $I[i, j]$ is the magnitude of the pixel at coordinates (i, j) ; Number of rows = r ; Number of columns = c).

- Classify noise type based on thresholds in Train images.

3. For each test image:

```

z_score = (mean_magnitude - avg_mean_magnitude)/std_dev
if z_score < threshold:
    return 'Gaussian Blur Noise'
elif z_score > threshold:
    return 'Impulse Noise'
else: return 'No Noise'
  
```

4. Compare with actual and predicted noise for each image.

5. Calculate accuracy for each noise type:

- Accuracy = (Correct Predictions/Total Predictions) * 100.

6. Print accuracies for Gaussian Blur, Impulse Noise, and No Noise.

For the examination of noise patterns in images, statistical techniques like mean and standard deviation are employed. The frequency domain approach is used for the observation of noise patterns. Three color channels are provided for input images; to convert an image to grayscale, take the mean along the channel axis. Calculated statistical measures for each form of image show the frequency domain properties

of various image types and noise patterns, and threshold values are produced based on these measurements. The training dataset's mean and standard deviation of the mean magnitudes for each category of noise are used to calculate these thresholds.

After that, determine the noise type next using the z-score. The z-score is a statistical approach for binary categorization. It quantifies how far the image's mean magnitude differs from the training dataset's mean magnitude. Gaussian blur noise often reduces an image's high-frequency components, resulting in a smoother frequency range and lower mean magnitude values. Impulse noise affects the smooth transition between frequencies in the frequency domain, resulting in unpredictable spikes or rapid intensity changes and higher mean magnitude values. The z-score determines the outcome; if it is less than -1.0 , there is "Gaussian blur noise" in the image. The image is classified as "impulse noise" if the z-score is more than -1.0 , and "no noise" if the z-score is between -1.0 and 1.0 .

5 Result

In the training process, more than 300 images were used. During testing, 40 images were utilized. Table 1 shows the outcome of 40 test images with their "actual noise" type and "predicted noise" type.

Table 1 shows noise classification results for various test images, showing both correct and incorrect predictions made by the frequency domain analysis. It provides a list of image names along with "predicted noise" type.

Accuracy results are calculated given formula, which is mentioned in Eq. (1).

$$\text{Accuracy} = (\sum C / \sum T) * 100 \quad (1)$$

where C = Number of correct predictions, T = Total number of predictions.

Here, correct predictions are the count of instances where the model's predictions matched the true labels. Total number of predictions is the total number of images which were used in testing.

Table 2 provides a clear overview of the accuracy percentages for each type of noise classification. The difference of correct predictions and the entire sample size of images is shown in the bar chart in Fig. 3.

The model's performance appears to be best at identifying Gaussian blur noise, followed by noise-free images and then impulse noise.

Table 1 Comparison of actual noises and predicted noises with different image samples

Gaussian blur detection		Impulse noise detection		No noise detection	
Test image name	Predicted noise	Test image name	Predicted noise	Test image name	Predicted noise
I01_01_05.png	Gaussian blur	I03_13_05.png	Impulse noise	I16.png	No noise
I02_01_05.png	Gaussian blur	I05_13_05.png	Impulse noise	I22.png	No noise
I06_01_05.png	Gaussian blur	I20_13_05.png	No noise	I25.png	Gaussian blur
I16_01_02.png	No noise	I24_13_05.png	No noise	I30.png	Gaussian blur
I16_01_01.png	No noise	I30_13_05.png	Gaussian blur	I31.png	Impulse Noise
I16_01_03.png	No noise	I30_13_04.png	Gaussian blur	I32.png	Gaussian blur
I16_01_05.png	Gaussian blur	I35_13_05.png	Impulse noise	I71.png	No noise
I16_01_04.png	Gaussian blur	I39_13_05.png	Impulse noise	I33.png	No noise
I21_01_05.png	Gaussian blur	I40_13_05.png	Impulse noise	I41.png	No noise
I22_01_04.png	Gaussian blur			I42.png	No noise
I24_01_05.png	Gaussian blur			I57.png	No noise
I30_01_05.png	Gaussian blur			I66.png	No noise
I31_01_05.png	Gaussian blur			I68.png	No noise
I32_01_05.png	Gaussian blur				
I40_01_05.png	Gaussian blur				
I42_01_05.png	Gaussian blur				

Table 2 Prediction outcome of test images

Number of images	Number of total images	Number of truly predicted images	Accuracy (%)
Noise-free	15	9	60.00
Gaussian blur	16	13	81.25
Impulse images	9	5	55.56

6 Conclusion

In this paper, we propose an efficient image noise type classification model using frequency domain analysis. Our noise classification method identifies whether an image has Gaussian blur noise or impulse type noise. Experimental results show that our noise classification model achieves a good accuracy percentage in Gaussian noise type images which is 81.25% while impulse noise type classification gives average

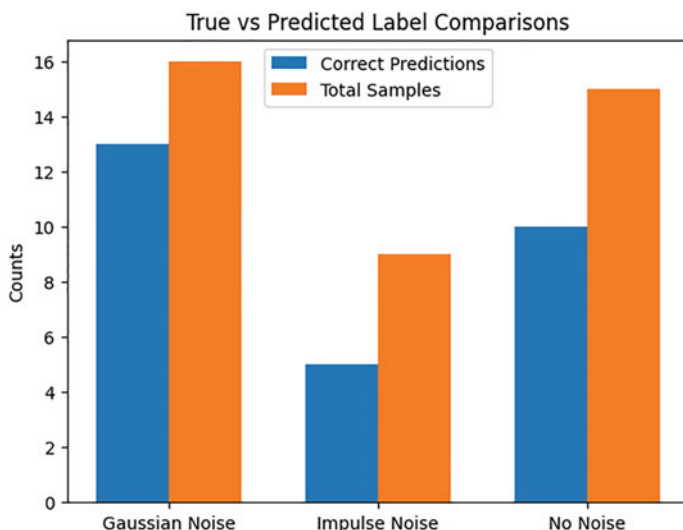


Fig. 3 Comparison of correct and predicted samples

performance with 55.56%. In future, the researchers intend to create a model that would be able to identify maximum type of noises within an image.

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Predictive Paradigm: AI-Driven Social Media Analysis for Real Estate Sales Forecasts



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Abstract Forecasting sales holds significant importance in the realms of production and supply chain management. It plays a crucial role in shaping a firm's planning, strategy formulation, marketing endeavors, logistical operations, warehouse management, and overall resource allocation. This paper employs five distinct machine learning (ML) algorithms, i.e., decision tree, random forest, K-nearest neighbors (KNN), K-means, and support vector machine (SVM) to forecast the accuracy percentage of real estate trends on social media platforms. A standard dataset comprising 200 respondents spanning various age groups is utilized. The simulation outcomes encompass predictions of accuracy percentages for heightened awareness, enhanced engagement, accelerated sales, and improved customer services.

Keywords Sales forecasting · Machine learning · Real estate companies · Warehouse management

1 Introduction

Sales forecasting holds a pivotal role in the operations of production, transportation, and decision-making across all echelons of a company's supply chain, as highlighted by Verstraete et al. [1]. The significance of sales forecasting has been a long-standing consensus among both scholars and industry practitioners, with Makridakis [2] and Remus and Simkin [3] being among the notable proponents. To delve further into its importance, these predictions form the bedrock of an organization's planning and strategic initiative [4], the allocation of corporate resources [5], marketing endeavors, and the efficient management of logistics and supply chain activities [6].

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The value of forecasts has surged significantly due to truncated lead times, heightened customer expectations, and the necessity to navigate within the confines of limited resources [7]. Proficient demand forecasting generally results in decreased inventory levels and enhanced customer service [8]. Accurate forecasting also assumes a critical role in retail and manufacturing operations [9]. In certain industries, forecasting serves as a linchpin for myriad operational and strategic decisions that directly affect a firm's profitability. For instance, in the domain of production planning, long-term forecasting is employed to ascertain the appropriate manpower levels and serves as a cornerstone for business planning, encompassing considerations for the expansion or contraction of production units [10].

Typically, when a company embarks on the implementation of a forecasting system, there exists a tendency to replicate concepts, objectives, and principles borrowed from other companies to expedite the process. Given that the majority of forecasting methods have been tailored for consumer products, there is a risk of adopting unrealistic targets and ill-suited error metrics when applied in different contexts [11]. The absence of forecasts or the presence of inaccuracies in forecasting can lead to the loss of sales or an excessive inventory surplus, thereby imposing additional costs on the firm [12]. For instance, imprecise demand forecasts have been responsible for sub-optimal levels of production workforce, which in turn result in workload imbalances and an escalation in the expenses related to recruitment, layoffs, and overtime labor activities. In numerous instances, relying on outsourced material supply in tandem with flawed forecasts can culminate in a dearth of supplies for production [13].

Consequently, swift responses to sales outcomes complemented by a well-managed distribution of forecasting errors contribute significantly to mitigating demand-related risks and maintaining equilibrium between production materials and inventory levels. Such responses acquire heightened significance in times of disasters, unforeseeable incidents, and volatile market conditions.

2 Related Works

Research on ESF [14] is rather scarce, and the predominant emphasis within existing studies related to ESF pertains to enhancing export forecasts, refining the precision of export sales forecasts, and optimizing overall performance in this domain. In the past, the primary focus of forecasting research was to identify the most suitable model for a given dataset. Regardless of the specific context, a range of methods has traditionally been employed for sales forecasting. Typically, demand and sales forecasting have relied on conventional time series techniques such as the ARIMA model and the Holt–Winters approach. However, there has been a growing interest in newer AI-based approaches due to their ability to enhance predictive accuracy and capture nonlinear patterns, as highlighted by Karmy and Maldonado [15]. Das et al. [16] contend that traditional statistical forecasting methods rely on extrapolating historical trends and seasonal fluctuations to predict the future. Consequently, these

methods often fall short when it comes to predicting significant environmental and macroeconomic changes in the business landscape, which can have a substantial impact on demand.

Winklhofer and Diamantopoulos [17] conducted research to examine a path model of ESF behavior and performance in order to identify the primary factors influencing the accuracy of export sales forecasting. Keck et al. [18] demonstrated that employing standard time series methods can significantly improve OECD25 trade forecasts compared to naive predictions. Furthermore, Ca'Zorzi and Schnatz [19] made forecasts for extra Euro-area exports using various measures of price and cost competitiveness. They determined that, in the case of a recursive estimation approach, the effective exchange rate based on the export price index outperforms other measures.

3 Methodology

In this section, we have applied different machine learning (ML) models such as decision tree (DT), random forest (RF), K-means, and K-nearest neighbor (KNN) which have been illustrated in the subsequent subsections.

3.1 Machine Learning Models

1. **Decision Tree:** A decision tree is a versatile machine learning algorithm that represents a hierarchical structure for making decisions or predictions. It is constructed recursively by splitting the data at each node based on specific criteria, such as Gini impurity or mean squared error for classification and regression tasks, respectively. These splits create branches, and the process continues until a stopping criterion, such as a maximum depth or a minimum number of samples, is met. Each leaf node of the tree holds a prediction or class label, and when given new data, it traverses the tree's branches based on feature values to arrive at a final prediction. Decision trees are interpretable, easy to visualize, and serve as the foundation for more complex ensemble methods like random forests and gradient boosting. The mathematical expressions for calculating accuracy in DT are as follows:

$$\text{Gini}(p) = 1 - \sum (\pi^2) \quad (1)$$

for each class ' i ' in a node, where ' p ' represents the proportions of each class in the node.

2. **Random Forest:** Random forest is an ensemble learning algorithm renowned for its efficacy in predictive modeling. Comprising an ensemble of decision trees, each constructed from bootstrapped samples of the dataset, random forest introduces variability through both sample selection and feature considera-

tion. Notably, at each node of the decision trees, a random subset of features is evaluated for optimal splits, mitigating potential biases and enhancing model robustness. This amalgamation of diverse decision trees results in a robust, high-performing model. In classification tasks, the final prediction is determined through a majority vote among the individual trees, while regression tasks involve averaging the predictions across the ensemble. The ensemble nature of random forest not only bolsters predictive accuracy but also furnishes resilience against overfitting. This algorithmic approach, blending randomness with ensemble principles, has found widespread application in various domains due to its ability to deliver superior predictive performance and robustness in the face of complex datasets and high-dimensional feature spaces.

Certainly, let us delve into the mathematical expressions underlying the random forest algorithm. Consider a dataset D comprising N instances, each with M features, denoted as $X = \{x_1, x_2, \dots, x_N\}$ and their corresponding labels $Y = \{y_1, y_2, \dots, y_N\}$. The random forest ensemble is constituted by T decision trees, each constructed from a bootstrapped sample of the dataset D_i and considering a random subset of features at each split. The construction of the t th tree is encapsulated by the recursive decision tree function $h_t(x)$, resulting in an ensemble $H(X) = \{h_1(x), h_2(x), \dots, h_T(x)\}$. For classification tasks, the final prediction is obtained through majority voting:

$$C_{\text{final}} = \operatorname{argmax}_c \sum_{t=1}^T I(h_t(x) = c) \quad (2)$$

where $I(\cdot)$ is the indicator function. In regression tasks, the ensemble prediction is the average of individual tree predictions:

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (3)$$

This ensemble approach, incorporating randomness through bootstrapped samples and feature subsets, contributes to the algorithm's predictive power and resilience.

3. **K-Nearest Neighbor:** The K-nearest neighbors (KNN) algorithm, a fundamental technique in machine learning, operates on the principle of proximity-based classification. Given a dataset of instances in a high-dimensional feature space, KNN classifies a new data point by examining the class labels of its k nearest neighbors. The choice of proximity metric, such as Euclidean or Manhattan distance, is pivotal in determining neighbor closeness. This algorithm is inherently nonparametric and exhibits flexibility in handling complex decision boundaries. However, its computational efficiency is contingent on the dataset's dimensionality and size. The KNN approach, although simple in concept, bears significance in applications ranging from pattern recognition to bioinformatics. Its efficacy is contingent on appropriate parameter selection, such as the choice of k , and

considerations for feature scaling to ensure uniform influence across dimensions. This algorithmic paradigm, rooted in geometric intuition, navigates the intricate terrain of high-dimensional spaces, thereby contributing to its enduring relevance in diverse domains.

Certainly, let us formalize the mathematical expression for the K-nearest neighbors (KNN) algorithm. Consider a dataset D comprising N instances, each with M features, denoted as $X = \{x_1, x_2, \dots, x_N\}$ and their corresponding class labels $Y = \{y_1, y_2, \dots, y_N\}$. Given a new data point x_q , the KNN algorithm determines its class label by identifying its k nearest neighbors based on a chosen proximity metric, such as Euclidean distance. The class C_q of the new data point is then determined by a majority vote among its k neighbors:

$$C_q = \operatorname{argmax}_c \sum_{i=1}^k I(y_i = c) \quad (4)$$

where $I(\cdot)$ is the indicator function. The flexibility of KNN lies in its simplicity and reliance on local information, making it particularly effective for nonparametric classification tasks where the decision boundaries are intricate and nonlinear.

4. **K-Means Algorithm:** The K-means algorithm, a foundational clustering technique in unsupervised machine learning, endeavors to partition a given dataset into k distinct clusters based on feature similarity. Operating under the assumption that each cluster is characterized by a centroid representing the mean of its constituent data points, K-means iteratively refines these centroids to minimize the within-cluster sum of squared distances. The algorithm's convergence is achieved through an alternating optimization process, alternating between the assignment of data points to the nearest centroid and recalibration of cluster centroids. While computationally efficient, K-means is sensitive to initial centroid placements, requiring careful consideration of initialization strategies. Its application spans diverse domains, from image segmentation to customer segmentation, owing to its ability to uncover underlying patterns within unlabeled data. Despite its widespread adoption, K-means is not without limitations, particularly in handling non-spherical clusters and accommodating varying cluster sizes. Addressing these challenges, variations of the algorithm and hybrid approaches have been proposed to enhance its robustness and applicability across diverse datasets. The key mathematical steps of the K-means algorithm are briefly given:

- (1) Initialization: Select k initial centroids: $\mu_1^{(0)}, \mu_2^{(0)}, \dots, \mu_k^{(0)}$.
- (2) Assignment Step: Assign each data point x_i to the nearest centroid: $c_i = \operatorname{argmin}_j \|x_i - \mu_j^{(t)}\|^2$.
- (3) Update Step: Update centroids by computing the mean of assigned data points for each cluster: $\mu_j^{(t+1)} = \frac{1}{|C_j|} \sum_{i \in C_j} x_i$.
- (4) Convergence: Repeat assignment and update steps until convergence, typically defined by a negligible change in centroids.

5. **Support Vector Machine:** The support vector machine (SVM), a powerful supervised learning algorithm, seeks to discern optimal hyperplanes that distinctly separate different classes within a high-dimensional feature space. By identifying the hyperplane that maximizes the margin, defined as the distance between the hyperplane and the nearest data points from each class, SVM achieves robust classification. The algorithm accommodates nonlinearly separable datasets by employing kernel functions to map data points into a higher-dimensional space, facilitating linear separation. Additionally, SVM excels in handling high-dimensional datasets with a limited number of instances. Its formulation involves the optimization of a convex quadratic programming problem, making it computationally efficient. SVM's versatility extends beyond binary classification to multi-class scenarios through techniques like one-versus-one or one-versus-all. Despite its effectiveness, the performance of SVM is contingent on appropriate parameter tuning, such as the choice of the kernel function and regularization parameters. Consequently, SVM stands as a formidable tool in various domains, including image recognition, bioinformatics, and text classification, owing to its capacity to discern intricate decision boundaries within complex datasets.

4 Experimental Analysis and Discussion

4.1 Data Collection

To gather data within the real estate industry, various questionnaires were developed, covering aspects like personal details, age, social media presence, and the perceived impact of social media on real estate companies. The respondents were then presented with comparative questions, prompting them to rate the effectiveness on a scale of 1 (not effective at all) to 5 (extremely effective). These questions gaged opinions on the success of social media in improving sales, enhancing customer service, increasing awareness, and improving engagement in real estate projects as shown in Fig. 1 by considering the mix of areas within Mumbai. The dataset comprises the city's diverse demographics where occupations could range from professionals in the real estate sector (like agents, brokers, and developers) to individuals in other industries who are also involved in property transactions. The dataset has been categorized into five multi-class classifications. To predict the influence of social media on the real estate industry, five distinct machine learning algorithms—decision trees (DT), random forest (RF), K-nearest neighbors (KNN), K-means, and support vector machine (SVM)—were applied.

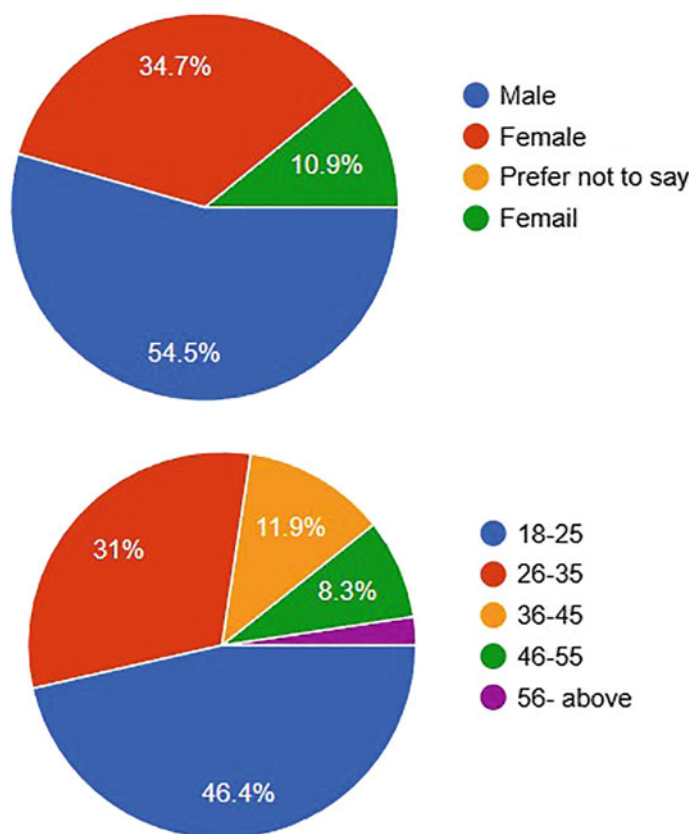


Fig. 1 Data collection from respondents

4.2 Simulation Results

The Python Integrated Development Environment (IDE) platform served as the foundation for various simulation tasks. The execution of diverse machine learning algorithms, including decision trees (DT), random forest (RF), support vector machine (SVM), K-nearest neighbors (KNN), and K-means, was employed to predict the accuracy of social media platforms' impact on the real estate industry. Specifically, social media channels such as Facebook, Instagram, Twitter, and LinkedIn were chosen for advertising real estate enterprises, as depicted in Fig. 2.

Nonetheless, crucial elements like heightened awareness, enhanced engagement, accelerated sales, and improved customer services have been identified as pivotal factors for advertising real estate companies. The diverse proportions are calculated from the above factors, as indicated by the collected responses. The prediction accuracy of sales has been computed for five distinct machine learning algorithms concerning the success of social media in real estate projects, as illustrated in Fig. 3.

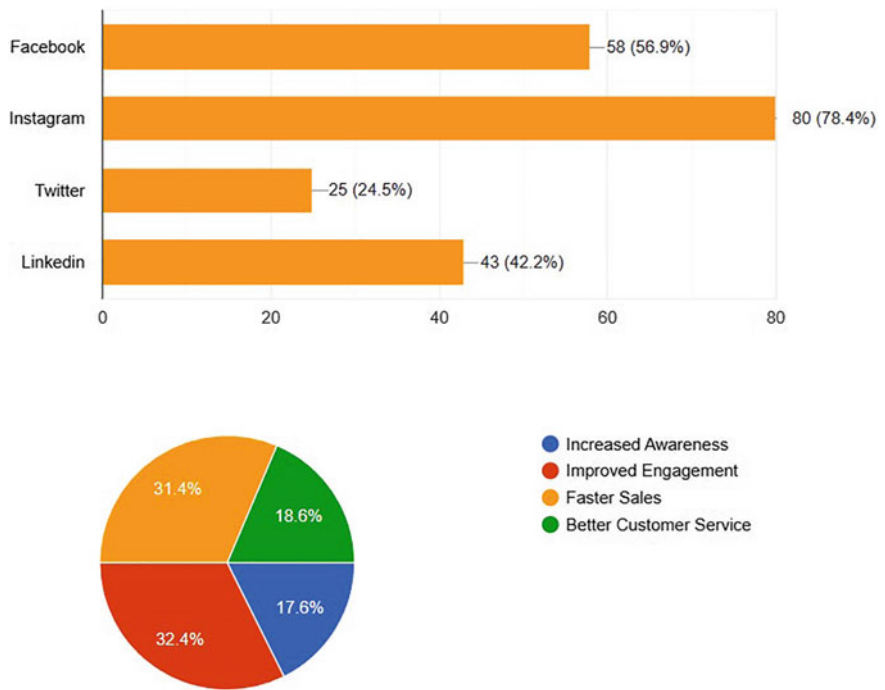
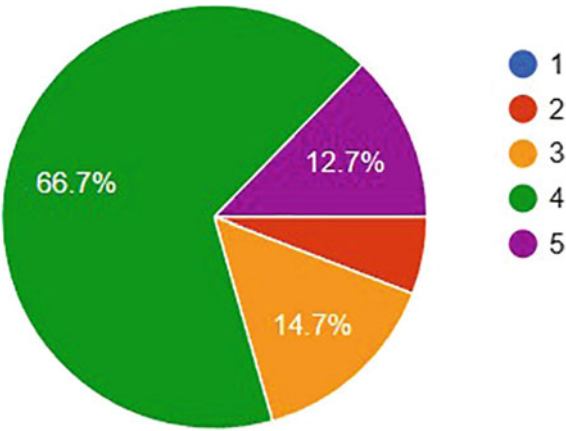


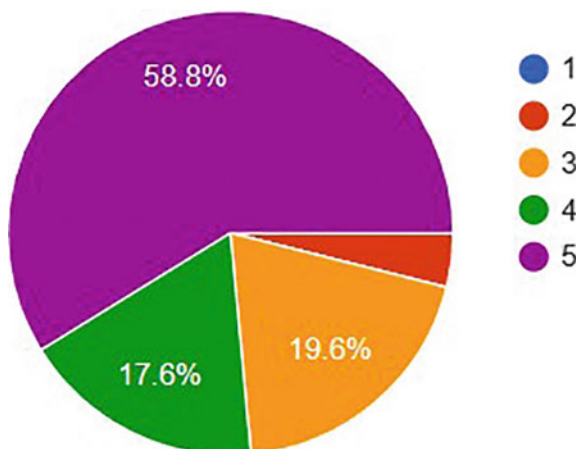
Fig. 2 Percentage of data collection from various social media sources

Fig. 3 Percentage prediction of sales by four machine learning algorithms



Various prediction accuracies of customer services have been computed in the context of real estate companies, as detailed in Fig. 4.

Fig. 4 Percentage prediction of customer services by four machine learning algorithms



5 Conclusion

In conclusion, sales forecasting emerges as a pivotal factor influencing various facets of a firm's operations, from strategic planning to resource allocation. The utilization of five diverse machine learning algorithms, including decision tree, random forest, K-nearest neighbors (KNN), K-means, and support vector machine (SVM), in predicting the accuracy of real estate trends on social media platforms showcases the potential of advanced analytics in this domain. The generalizability of the findings may be limited to the specific context or region under study. Future research could aim to validate the robustness of the proposed models across diverse geographical locations and real estate markets. The study, based on a dataset featuring 200 respondents across different age groups, contributes insights into forecasting accuracy for key aspects such as heightened awareness, enhanced engagement, accelerated sales, and improved customer services. Looking ahead, the future scope lies in refining these predictive models, exploring more advanced ML techniques, and incorporating real-time data to enhance the precision of sales forecasts in dynamic market environments. Additionally, the integration of interdisciplinary approaches and continuous adaptation to evolving technologies will further elevate the effectiveness of sales forecasting in the ever-changing landscape of production and supply chain management.

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